

COURSE DESCRIPTION			
Program	Artificial Intelligence, Artificial Intelligence & Machine Learning, Cyber Forensics and Information Security, Computer Science and Technology, Computer Science & Engineering, Computer Engineering, Computer Science and Engineering (Artificial Intelligence & Machine Learning), Information Technology, Robotic Process Automation		
Course Title	Programming in C		
Course Code	2132	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces the fundamental concepts of programming using the C language. It focuses on problem-solving techniques, structured programming concepts, control structures, functions, arrays, pointers, and basic data structures. The course enables students to develop logical thinking and implement efficient programs using C, which forms the foundation for advanced programming and core computer engineering subjects.

Course Objectives

1	To enable learners to apply logical thinking and problem-solving techniques to solve real-world problems.
2	To equip learners with the ability to write efficient and structured C programs using appropriate language constructs for real-world computational problems.
3	To provide learners with a strong foundation in programming concepts, enabling them to understand and learn advanced programming languages.
4	To familiarize learners with basic data structures and memory concepts, preparing them for further studies in computer science and engineering.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in problem solving and programming concepts.	1131	Problem solving and Python programming	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Make use of the basic programming concepts -sequential, conditional, looping structures in C.	Apply
CO2	To develop programs using array, matrices and string	Apply
CO3	To develop problem solving skills in modular programming using functions and structure	Apply
CO4	Study the concepts of Pointer, Files, Dynamic memory allocation and develop programs using pointers and files	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3	3	3	-	-	2	-	-	-	2
CO2	3	3	3	3	-	-	2	-	-	-	2
CO3	3	3	3	3	-	-	2	-	-	-	2
CO4	3	3	3	3	-	-	2	-	-	-	2
Total	11	12	12	12	-	-	8	-	-	-	8
Correlation Level	2.75	3	3	3	-	-	2	-	-	-	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Basics of C Programming & Control Statements	C character set, Data types, Operators, Structure of C program, I/O statements, Conditional control statements, Iterative control statements, Control transfer statements.	17	0
Module II	Arrays and Strings	Arrays, Single Dimensional array, Multi dimensional array, Strings	15	0
Module III	Functions and Structures	Functions, Recursion, Structures, Unions	15	0
Module IV	Pointers and File Handling	Pointers, File operations	13	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Basic concepts in C language	Define Character Set, Constants, Variables, Identifiers, Keywords, Token, Data type. Illustrate variable initialisation and declaration.	L	CO1	Understand	2
Operators and its Precedence, Expression.	Explain different Operations in C- Arithmetic, Relational, logical, Conditional, Unary, Increment and Decrement, Assignment operators.	L	CO1	Understand	2
Structure of C programs.	Define Input and output Statements -printf() and scanf() with format specifier. Explain the Structure of C programs. Illustrate Simple C programs using input, output functions.	L	CO1	Apply	3
Control Statements (Decision Control Statements)	Use if, if-else, nested if, switch control statements to solve problems.	L	CO1	Apply	4
Iterative Control Statements	Use while, do-while, for loop, nested loops to execute statements repeatedly in a program.	L	CO1	Apply	5
Control Transfer Statements	Define and Use break and continue statements	L	CO1	Apply	1
Module II					
Array	Define Array. List different types of Array- Single dimensional Array. Multi dimensional Array. List Applications of Array - Searching, Sorting and Matrix Representation. Define Single dimensional array- Declaration, Initialization, Accessing and manipulating array elements. Solve different simple applications using Array.	L	CO2	Apply	5
Array	Define Two dimensional array, Declaration, Initialization, Accessing and manipulating array elements. Solve different matrix applications using Array.	L	CO2	Apply	5
Strings	Define Strings - Declaring a string variable, Reading and displaying strings, String related programs using library functions.	L	CO2	Apply	5
Module III					

Functions	Functions - Definition and purpose, Types of functions- user defined and built in functions. Components of Function with its syntax(Function definition, Function call, Function prototype), Parameter passing- Pass by value and Pass by reference Passing Array to a Function. Solve different simple applications using function.	L	CO3	Apply	7
Recursion	Explain Recursion. Develop program using recursion	L	CO3	Apply	2
Structure	Understand the concepts of Structure- Definition, Declaration, Accessing members. Understand Union- Definition, Declaration syntax, Difference between Union and Structure.	L	CO3	Understand	6
Module IV					
Pointers	Explain pointers, its declaration. Passing a pointer to a function. Solve problems using pointers.	L	CO4	Apply	6
Files	Define File, Different types of files in C, modes of file operations, Opening & Closing a file, Read and write operation in file. Solve different operations on text files.	L	CO4	Apply	6

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare a table which consists of columns with real-world- data, datatype, variable- name, declaration- in- C. Fill up the correct data type of that data, use a valid variable name and write its declaration against the real-world-data given.	2
1	Familiarise the execution steps in C program and Online C compilers.	2
1	Given Quiz related to basic programming concepts in C using google form	2
2	Write algorithm for Print numbers from 1 to 100, Find factorial of a number, Generate Fibonacci series	3
2	Write 5 simple program in online C compiler and submit the output screenshot	3
3	Provide code snippet and predict the output and errors to understand C programming concepts	3
3	Write and submit the output screenshot to google classroom 1. Check if a number is even or odd 2. Find the largest of three numbers 3. Display multiplication table of any number	3
Module II		
4	Write programs to find sum, average, maximum, and minimum of array elements and submit the output screenshot to google classroom	3
4	Draw array memory organization for a list of values in one dimensional and two dimensional array and justify the differences.	
5	Write any three string handling function program and do the same without using string handling function.	3
5	Show by drawing how Sorting performed for a list of values using bubble sort in each iteration	3
6	Show by drawing how linear search works in a list of values for a key value in each iteration	2
6	Show how two matrices get added by your Matrix addition program	2
6	A quiz in any of the online mode about array concepts	2
Module III		
7	Write any three program using function with call by value and call by reference	3
7	Give a code snippet using a function to predict the output.	3
8	Understand the concept of recursion with any example program	6
9	Explain how arrays of structures are useful in solving programs. Justify your answer with an example.	3
9	Describe the usage of storage class specifiers. Specify its scope, visibility in a program with examples.	3
Module IV		
10	Find and implement an application that passes pointers as function arguments.	6
11	Create a table listing: File modes (r, w, a, r+, w+, a+) write Purpose of each mode	6
12	Provide incorrect code (missing fclose(), wrong mode). Students identify and correct errors.	6
13	Submit your findings about why FILE * a pointer? What is the difference between normal pointer and file pointer	6
14	Given a pointer-based program, students will be able to Trace pointer values and write memory addresses and values step-by-step.	6

15	Write C program to perform file operations and submit the output in google classroom	3
15	Conduct a quiz in google form	3
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Programming in ANSI C	E. Balagurusamy	McGraw-Hill
2	The C Programming Language	Kernighan & Ritchie	Pearson

REFERENCE			
Sl. No.	Title	Author	Publication
1	C Programming Absolute Beginner's Guide	Greg Perry	Pearson
2	Let Us C	Yashavant Kanetkar	BPB

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.geeksforgeeks.org/c-programming-language/	All
2	https://www.tutorialspoint.com/cprogramming	All

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	

2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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